Terrain-Aware A* with Robot-Grounded Traversability Costs: A Reproducible Evaluation for Tracked UGVs on Uneven Terrain

Author Name
Affiliation
City, Country
email@example.com

***Abstract*—Shortest-path planners routinely send ground robots across slopes and rough patches that a cost-aware planner would avoid at small extra distance. We study a simple and fully specified terrain-aware planner for tracked unmanned ground vehicles (UGVs): a traversability cost that is normalised by the robot's own limits (slope limit and roughness saturation), embedded in A* as a multiplicative edge cost with an admissible heuristic (TA*), followed by a shortcutting step that provably never increases the integrated cost (TA*-S). Our main contribution is a careful, reproducible evaluation: 120 test scenarios over four procedurally generated terrain families and three map scales, a held-out validation split used only to select the cost weights, planner-independent safety and energy metrics, and paired non-parametric statistics. Against terrain-unaware A* and Theta*, TA*-S changes the median traversability cost along the path by -39.6% relative to A* and never enters terrain steeper than the slope limit, whereas A* and Theta* do so in 76% and 82% of the scenarios; the median path-length change is +3.0% and the median change in estimated energy is -7.3%. Compared with a binary-traversability baseline (hard slope constraint only), the continuous cost further lowers the slope, roughness and climb along the path (all $p < 0.001$, Holm-corrected). A heading-dependent extension that separates pitch from roll keeps every path within an assumed roll limit of 20°, which shortest-path planners violate in 71% of the scenarios. We also report a joint optimisation of the cost weights, a cost ablation, infeasible maps, robustness to elevation-map noise and computation time. Code and data generators are released.**

***Index Terms*—path planning, traversability, field robotics, unmanned ground vehicles, uneven terrain**

I. Introduction

Field robots in agriculture, construction and disaster response operate on terrain where the shortest path is often not the best one: a straight line over a hill or through a rough patch costs energy, increases the risk of slipping or tipping, and may exceed the vehicle's climbing capability. Terrain-aware planning addresses this by attaching a traversability cost to the map and searching for paths that trade distance against terrain difficulty [1], [2].

The idea itself is well established—planetary rovers have used traversability maps for two decades [3], [4]—and adding a terrain term to a graph-search cost is, algorithmically, a small step. What is often less clear in practice is *how much* such a cost actually buys on a given robot, under which conditions, and at what price in distance, energy and computation. Answering this requires evaluations whose metrics are tied to the robot (e.g., distance driven on slopes beyond its limit), whose baselines include both terrain-unaware and binary traversability planners, whose parameters are not tuned on the test data, and whose statistics respect the paired design of planner benchmarks.

This paper provides such an evaluation for a tracked UGV. Our contributions are:

1) A fully specified terrain-aware planner: a slope/roughness cost normalised by the robot's limits rather than by per-map maxima, combined with hard constraints, used as a multiplicative edge cost in A* with an admissible heuristic (TA*), and a cost-aware shortcutting step (TA*-S) that never increases the integrated cost, with a heading-dependent extension that bounds roll as well as pitch (Sec. III).
2) A reproducible benchmark of 120 test and 60 validation scenarios over four terrain families and three scales, and a joint optimisation of the cost weights on the validation split only, with mission-specific selection rules and a stability analysis (Secs. IV, V-C).
3) An evaluation with planner-independent safety and energy metrics and paired, multiplicity-corrected statistics, including the distance–terrain trade-off, an ablation of the cost terms, robustness to elevation noise and computation time (Sec. V).

All numbers in this paper are generated from the released raw results by a single script.

II. Related Work

Grid search. A* [5] finds minimum-cost paths on grids given an admissible heuristic; D* Lite [6] supports efficient replanning. Any-angle planners such as Theta* [7], [8] and Field D* [4] remove the grid's heading restriction; Field D* additionally interpolates costs across cells and was deployed on Mars rovers.

Traversability. Surveys [1], [2] distinguish geometric approaches, which derive slope, roughness and step height from elevation maps or point clouds, from appearance- and learning-based ones. Elevation mapping with uncertainty [9] and SLAM systems such as RTAB-Map [10] provide the maps on which such features are computed. Geometric traversability costs have been used for legged robots [11], for planning directly on point clouds [12] and for energy-efficient paths on uneven terrain [13].

Positioning. We do not propose a new search algorithm. We contribute a carefully specified cost and planner for tracked UGVs and, mainly, an evaluation protocol whose metrics, baselines, data split and statistics make the benefit of terrain awareness measurable and reproducible.

III. Method

A. Elevation map and terrain features

The environment is a 2.5D elevation map $z(x, y)$ on a grid of resolution $\Delta = 0.5\,\mathrm{m}$, as produced, for example, from an RTAB-Map point cloud [10]. For each cell we compute the slope $\theta = \arctan \|\nabla z\|$ (central differences) and the roughness ρ, the local standard deviation (3×3 window) of the residual $z - G_\sigma * z$ after removing smooth relief with a Gaussian filter ($\sigma = 1.5$ cells). Geometric obstacles (rocks) are inflated by the robot's footprint radius r_f.

B. Robot-grounded cost and constraints

Let $\theta_{\max}$ be the robot's admissible slope and ρ_{sat} a roughness at which the roughness penalty saturates. The cost of a cell is

$$c = \alpha \min\Big(\frac{\theta}{\theta_{\max}}, 1\Big) + \beta \min\Big(\frac{\rho}{\rho_{\mathrm{sat}}}, 1\Big), \quad \alpha + \beta = 1, \quad (1)$$

so $c \in [0, 1]$. Because the normalisers are robot properties, the same terrain receives the same cost on every map; normalising by the per-map maximum, as is sometimes done, makes costs incomparable across maps and lets a single outlier rescale the whole map. Cells with $\theta > \theta_{\max}$ or inside an inflated obstacle are *lethal*. We use $\theta_{\max} = 25°$, $\rho_{\mathrm{sat}} = 0.10\,\mathrm{m}$ and $r_f = 0.5\,\mathrm{m}$. The weights (α, β) and the terrain weight w below are selected on validation data (Sec. IV-C).

C. Terrain-Aware A* (TA*)

The grid is 8-connected without corner cutting. The cost of moving between neighbouring free cells u, v at horizontal distance d_{uv} is

$$k(u, v) = d_{uv}\Big(1 + w\,\frac{c_u + c_v}{2}\Big), \quad (2)$$

with terrain weight $w \geq 0$. The path cost g accumulates k; the heuristic is the Euclidean distance to the goal.

Proposition 1. *For any $w \geq 0$ the Euclidean heuristic is consistent for* (2)*; hence TA* returns a minimum-cost path.*

Proof. Since $c \geq 0$, $k(u, v) \geq d_{uv} \geq h(u) - h(v)$ by the triangle inequality. Consistency implies optimality of A* [5]. □

Note that the terrain term must enter the *accumulated* cost g: adding it only to the priority $f = g + h + w\,c(n)$ changes the expansion order but not the objective that the returned path optimises. With $w = 0$, TA* reduces to A* with hard constraints (A*-HC), i.e., binary traversability.

D. Cost-aware shortcutting (TA*-S)

Grid paths contain heading artefacts. Standard string-pulling removes them but can cut straight across costly terrain. TA*-S shortcuts greedily: from waypoint p_i it extends a straight segment to the farthest p_j such that (i) the segment is collision-free with respect to lethal cells (a conservative visibility test that also checks cells touched at boundaries) and (ii) its integrated cost $\int (1 + w\,c)\,ds$ does not exceed the cost of the original sub-path $p_i \ldots p_j$ evaluated with the same quadrature.

Proposition 2. *TA*-S never increases the integrated path cost and never enters a lethal cell.*

Proof. Each accepted shortcut replaces a sub-path by a segment of no greater cost that passes the visibility test; the result is a concatenation of such segments. □

E. Roll-aware extension (TA*-S-R)

The cost (1) is isotropic: crossing a slope costs as much as climbing it, although crossing produces roll, which tracked vehicles often tolerate less than pitch. On a cell with elevation gradient ∇z, moving in unit direction $\mathbf{u}$ gives $\tan\psi = |\nabla z \cdot \mathbf{u}|$ (pitch) and $\tan\phi = |\nabla z \times \mathbf{u}|$ (roll). The heading-dependent variant replaces the slope term of (1) by $\min(\max(\psi/\theta_{\max},\ \phi/\phi_{\max}), 1)$, the larger utilisation of the two limits, and forbids moves with $\psi > \theta_{\max}$ or $\phi > \phi_{\max}$. No weight is added. Since pitch and roll depend only on the move direction, the 8-connected graph is kept, with one multiplier and one constraint mask per direction; costs stay $\geq$ the distance, so Proposition 1 still holds. Shortcutting uses the same directional cost and constraints. We call the result TA*-R and TA*-S-R. We use $\phi_{\max} = 20°$, which we consider representative of dry, loose sand: with the track half-width of our vehicle model (0.35 m) and an estimated centre-of-gravity height of about 0.34 m, static tip-over occurs only near $45°$, so lateral sliding and sinkage of the downhill track limit side slopes first. As this value is an assumption, it is varied between $10°$ and $25°$.

IV. Experimental Setup

A. Terrains and scenarios

We generate four terrain families (Fig. 1): *hills* (fractal relief, median slope $11°$), *ridge* (gentle relief with a steep ridge across the start–goal line and one or two passes), *craters* (gentle relief with 6–11 crater-like depressions) and *rocky* (gentle relief, rough patches and scattered rocks). Relief is produced by $1/f^\beta$ spectral synthesis and scaled to a target median slope, so slope statistics are comparable across the three map sizes (50, 100 and 200 m, i.e., up to 400×400 cells). A centimetre-scale micro relief is added everywhere. The start lies in the left and the goal in the right band of the map. A

scenario is kept only if a path exists under the hard constraints, so every planner faces a solvable instance; 1 of the 120 test scenarios needed a redraw. Each (family, size) cell contains 10 test seeds; a disjoint set of seeds (60 scenarios) forms the validation split. All randomness is seeded and independent of the Python hash seed.

B. Planners

A* (shortest path, obstacles only), Theta* (Lazy Theta* [8], any-angle shortest path, obstacles only), A*-HC (shortest path with hard constraints), TA* and TA*-S. All planners share the same grid, neighbourhood, data structures and language (Python 3, single thread), so their computation times are comparable to each other but not to optimised C++ implementations.

C. Weight selection

Since $\alpha + \beta = 1$, the pair (w, α) sets the slope weight $w\alpha$ and the roughness weight $w\beta$ independently. We evaluate TA*-S on a grid $w \in \{0, 0.5, 1, 2, 4, 8, 16, 32, 64\}$, $\alpha \in \{0, 0.25, 0.5, 0.75, 1\}$ (41 distinct configurations) on the 60 validation scenarios and select weights for three mission profiles, using objectives that do not depend on the weights themselves: *energy*, the minimum median energy change relative to A*; *safety*, the minimum median 95th-percentile slope under the path subject to a median length increase of at most 10%; and *balanced*, the knee of the energy–slope Pareto front, i.e., the front point closest to the utopia point after min–max normalisation. The balanced weights, $w = 8$ and $\alpha = 0.5$, are used in all other experiments. The test split is never used for selection.

D. Metrics

All paths are resampled every 0.1 m and evaluated on the ground-truth map with the same code: horizontal length; cumulative climb (sum of positive elevation changes); maximum and 95th-percentile slope under the path; *steep length*, the distance driven on cells with $\theta > \theta_{\max}$; mean cost $\bar{c}$ (1); 95th-percentile roughness; and an energy estimate $E = mg(\mu_r L + H^+)$ with $m = 150$ kg, rolling-resistance coefficient $\mu_r = 0.15$, length L and climb H^+ (no regeneration). The energy model is deliberately simple and is used only for relative comparisons. For the roll-aware extension we also measure the roll angle along the direction of travel and the distance driven with roll above $\phi_{\max}$.

E. Statistics

Planners are compared on the same scenarios, so we use paired two-sided Wilcoxon signed-rank tests [14], report the median paired difference and the matched-pairs rank-biserial correlation r [15] as effect size, and correct all 24 tests with Holm's method [16]. Distributions are summarised by median and interquartile range (IQR) because computation times and several metrics are strongly skewed.

V. Results

A. Main comparison

Table I summarises the 120 test scenarios and Table II the paired tests. All planners solve every scenario; since the benchmark contains only maps with a feasible path, this is expected, and infeasible maps are treated separately below. The terrain-unaware planners A* and Theta* enter terrain steeper than $\theta_{\max}$ in 76% and 82% of the scenarios, driving on average 8.5 m and 8.1 m on such slopes. A*-HC avoids lethal cells by construction but hugs the limit: its median maximum slope is 23.7° against $\theta_{\max} = 25°$. TA*-S keeps a margin: the median maximum slope under the path is 13.7° and the median maximum pitch over 1 m windows is 11.5° (A*: 26.8°, A*-HC: 19.0°).

Relative to A* on the same scenario, TA*-S changes the mean traversability cost by -39.6%, the cumulative climb by -39.3% and the estimated energy by -7.3% (median), at a median path-length increase of 3.0%. Against A*-HC, the continuous cost changes the 95th-percentile slope by -8.70° and the climb by -0.81 m (median paired differences, $|r| \geq 0.98$). Against Theta*, which produces the shortest paths, TA*-S is longer by a median of +6.59 m and still uses slightly less energy (median -0.21 kJ, $r = -0.32$, $p = 0.002$): the extra distance is more than paid for by less climbing. The shortcutting step matters: compared with plain TA*, TA*-S changes the length by -3.97 m (median) and uses less energy in every scenario ($r = -1.00$), at a small cost increase of +0.013.

B. Terrain families

Table III breaks the results down by terrain family. Exposure of the terrain-unaware planners is highest on the ridge terrain (97% of A* paths), where the shortest path crosses the ridge flank instead of using a pass. The energy benefit of TA*-S is largest on craters (-16.6%) and smallest on hills (-3.6%), where elevation changes cannot be avoided by short detours. On rocky terrain the roughness term dominates: the 95th-percentile roughness under the path is 37 mm for TA*-S versus 113 mm for A* and 104 mm for A*-HC.

C. Weight optimisation

Fig. 2 shows the validation grid. The terrain weight w controls a smooth trade-off: small weights minimise the energy estimate, large weights minimise the slope under the path, and no configuration is best for both. Mixing both cost terms pays off: the pure-roughness cost ($\alpha = 0$) is Pareto-dominated for every $w > 0$, and all three selected configurations use $\alpha \in \{0.5, 0.75\}$. Table IV reports the selected weights on the test split. The energy profile ($w = 1$, $\alpha = 0.75$) changes the energy by -9.5% and the length by -2.5% relative to A*, but leaves the 95th-percentile slope at 14.8°; the safety profile ($w = 32$, $\alpha = 0.75$) reaches 8.5° at +9.3% length; the balanced profile lies in between (-7.3% energy, +3.0% length, 9.3°). Even with the energy-oriented weights no test path enters terrain steeper than $\theta_{\max}$, because the hard constraint, not the weight, enforces the limit.

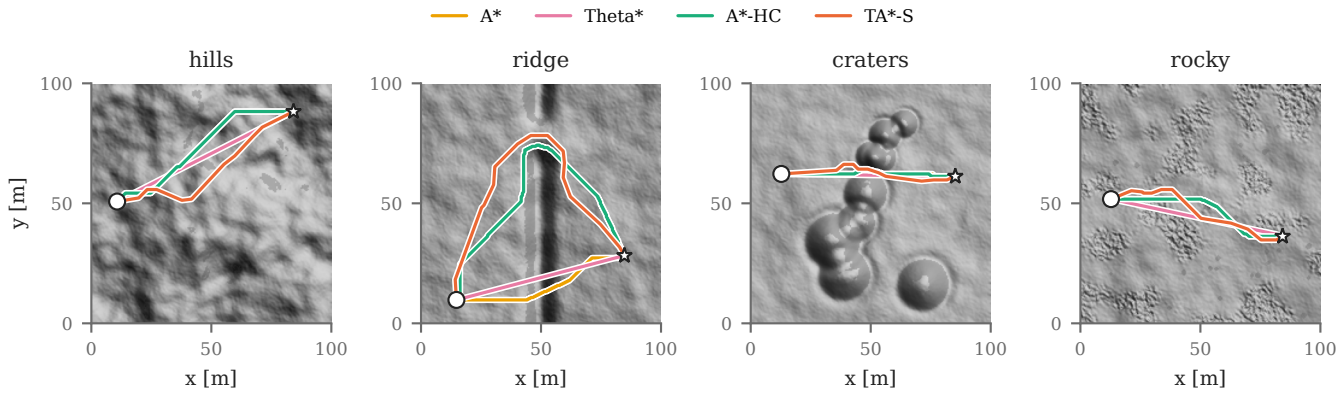


Fig. 1. Example test scenarios (100 m, seed 0) for the four terrain families on a hillshade; darker shading marks lethal cells (slope $> \theta_{\max}$ or inflated rocks). Circle: start, star: goal. Where lines are not visible they coincide with another planner's path (e.g., A* and A*-HC when no lethal cell lies on the shortest path).

TABLE I

RESULTS ON THE 120 TEST SCENARIOS. LENGTH AND ENERGY ARE RELATIVE TO A* ON THE SAME SCENARIO, MEDIAN [IQR]. MAX. SLOPE AND COST: MEDIANS. EXPOSED: SHARE OF PATHS THAT ENTER TERRAIN STEEPER THAN $\theta_{\max}$; STEEP LENGTH: MEAN LENGTH DRIVEN ON SUCH TERRAIN.

Planner	Succ. [%]	ΔLength [%]	ΔEnergy [%]	Max. slope [°]	Exposed [%]	Steep len. [m]	Cost $\bar{c}$	Time [s]
A*	100	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	31.1	76	8.5	0.300	0.021
Theta*	100	-5.7 [-7.0, -3.8]	-4.6 [-8.4, -1.5]	33.2	82	8.1	0.301	0.027
A*-HC	100	0.1 [0.0, 2.6]	-1.4 [-7.3, 0.0]	23.7	0	0.0	0.267	0.019
TA*	100	7.7 [4.4, 13.3]	-3.3 [-9.6, 1.3]	12.7	0	0.0	0.147	0.088
TA*-S	100	3.0 [-0.2, 8.0]	-7.3 [-13.2, -2.8]	13.7	0	0.0	0.160	0.116

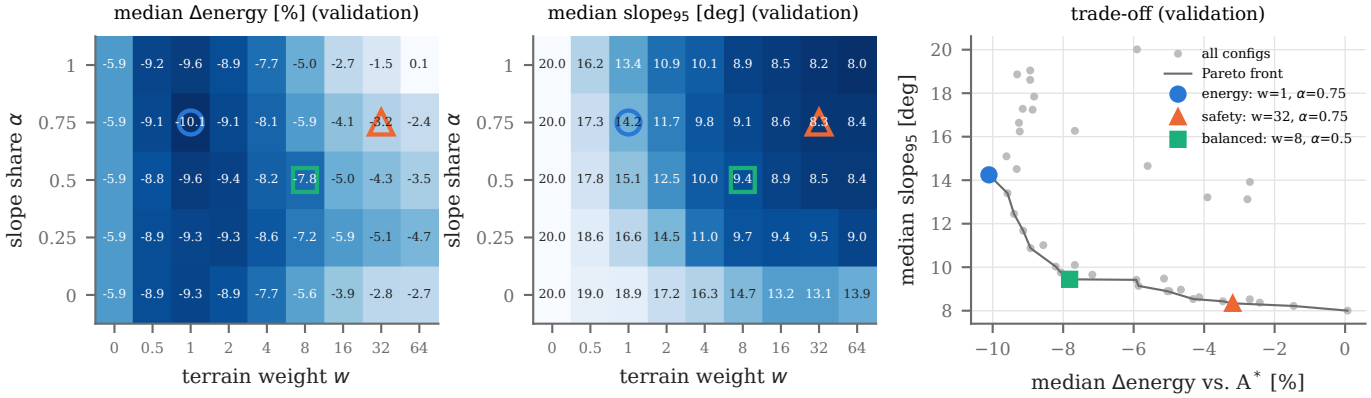


Fig. 2. Weight optimisation on the validation split. Left and centre: median energy change relative to A* and median 95th-percentile slope under the TA*-S path for every (w, α) ($w = 0$ ignores α). Right: all configurations, their Pareto front and the weights selected for the three mission profiles (circle: energy, triangle: safety, square: balanced).

The exact optimum is not stable: the balanced configuration is reselected in 46% of 300 bootstrap resamples of the validation scenarios and in 1 of 4 leave-one-terrain-family-out runs. The alternatives, however, are grid neighbours (a factor of two in w, $\alpha \in \{0.5, 0.75\}$) with nearly identical performance, so the objectives are flat around the optimum and the planner does not depend on precise tuning. The weight should be chosen per mission, coarsely.

D. Cost ablation

Table V varies the slope weight α ($\beta = 1-\alpha$). A roughness-only cost ($\alpha = 0$) leaves the 95th-percentile slope at 14.2°; any slope weight $\alpha \geq 0.25$ lowers it to 9.2° or less. Conversely, the 95th-percentile roughness grows from 18.4 mm to 22.6 mm as the roughness term is removed. The two terms thus act on the quantities they are designed for, and the selected $\alpha = 0.5$ lies on the flat part of the slope curve while retaining most of the roughness avoidance.

E. Maps without a feasible path

The benchmark keeps only solvable maps, so a success rate of 100% only means that every planner returned *some* path. To test the opposite case we generated 30 additional maps with a steep wall across the whole map and no pass (10 seeds per

TABLE II

PAIRED TWO-SIDED WILCOXON SIGNED-RANK TESTS OVER THE TEST SCENARIOS (N=120). $\tilde{d}$: MEDIAN PAIRED DIFFERENCE (FIRST MINUS SECOND); r: MATCHED-PAIRS RANK-BISERIAL CORRELATION; p: HOLM-CORRECTED OVER ALL 24 TESTS.

Comparison	Metric	$\tilde{d}$	r	p
TA*-S vs A*	cost $\bar{c}$	-0.106	-1.00	<0.001
	95th-pct. slope [°]	-12.28	-1.00	<0.001
	95th-pct. rough. [mm]	-8.4	-0.99	<0.001
	climb [m]	-1.69	-0.98	<0.001
	energy [kJ]	-1.85	-0.84	<0.001
	length [m]	+2.45	+0.65	<0.001
TA*-S vs Theta*	cost $\bar{c}$	-0.112	-1.00	<0.001
	95th-pct. slope [°]	-11.84	-1.00	<0.001
	95th-pct. rough. [mm]	-5.7	-0.98	<0.001
	climb [m]	-1.50	-0.94	<0.001
	energy [kJ]	-0.21	-0.32	0.002
	length [m]	+6.59	+1.00	<0.001
TA*-S vs A*-HC	cost $\bar{c}$	-0.091	-1.00	<0.001
	95th-pct. slope [°]	-8.70	-1.00	<0.001
	95th-pct. rough. [mm]	-4.8	-0.98	<0.001
	climb [m]	-0.81	-0.98	<0.001
	energy [kJ]	-1.08	-0.87	<0.001
	length [m]	+0.97	+0.42	<0.001
TA*-S vs TA*	cost $\bar{c}$	+0.013	+1.00	<0.001
	95th-pct. slope [°]	+0.51	+0.92	<0.001
	95th-pct. rough. [mm]	+0.9	+0.92	<0.001
	climb [m]	-0.05	-0.74	<0.001
	energy [kJ]	-0.97	-1.00	<0.001
	length [m]	-3.97	-1.00	<0.001

TABLE III

PER TERRAIN FAMILY (MEDIANS). $\bar{c}$: MEAN TRAVERSABILITY COST ALONG THE PATH; EXPOSED: SHARE OF PATHS ENTERING SLOPES $> \theta_{\max}$; Δ: TA*-S RELATIVE TO A*.

	$\bar{c}$				Exposed [%]		TA*-S Δ [%]	
Terrain	A*	Th*	HC	TA*-S	A*	Th*	Len.	Energy
hills	0.316	0.324	0.312	0.230	63	77	1.6	-3.6
ridge	0.245	0.244	0.220	0.148	97	90	4.8	-10.0
craters	0.320	0.312	0.246	0.143	83	80	3.5	-16.6
rocky	0.329	0.358	0.323	0.170	60	80	2.7	-5.8

map size), not filtered for feasibility. TA*, TA*-S and A*-HC correctly report that no admissible path exists in all of them (0, 0 and 0 paths returned), after at most 0.36 s. A* and Theta*, in contrast, "succeed" on all 30 maps (30 and 30 paths), but their paths cross the wall with a mean maximum slope of 48° and drive on average 16.3 m on slopes beyond the limit. For a terrain-unaware planner, success is therefore not evidence of a usable path.

F. Robustness to elevation noise

Real elevation maps are noisy. We add i.i.d. Gaussian noise $\sigma_z \in \{0.02, 0.05, 0.10\}$ m to the elevation used for planning and evaluate the resulting paths on the ground truth (Fig. 3). Noise inflates the estimated slope, so both planners occasionally misjudge cells near the limit and enter truly steep terrain. Because TA* keeps a margin from $\theta_{\max}$, its exposure stays smaller than that of A*-HC (mean 0.05 m vs. 0.10 m at $\sigma_z = 0.10$ m); both remain at or above 99.2% success. The mean cost of TA* paths rises from 0.157 to 0.209, i.e., the benefit degrades gracefully. Filtering the map before computing slopes, or using an uncertainty-aware elevation map [9], is the natural remedy.

TABLE IV

WEIGHTS SELECTED ON THE VALIDATION SPLIT FOR THREE MISSION PROFILES AND THEIR PERFORMANCE ON THE TEST SPLIT (MEDIANS; Δ RELATIVE TO A*). STABILITY: SHARE OF 300 BOOTSTRAP RESAMPLES OF THE VALIDATION SPLIT THAT SELECT THE SAME WEIGHTS / NUMBER OF LEAVE-ONE-TERRAIN-FAMILY-OUT SELECTIONS (OUT OF 4) THAT AGREE.

Profile	w	α	ΔE [%]	ΔL [%]	slope$_{95}$ [°]	rough.$_{95}$ [mm]	stab.
Energy	1	0.75	-9.5	-2.5	14.8	24.4	42%, 3/4
Safety	32	0.75	-2.9	9.3	8.5	21.4	29%, 1/4
Balanced	8	0.5	-7.3	3.0	9.3	21.6	46%, 1/4

TABLE V

ABLATION OF THE SLOPE/ROUGHNESS WEIGHTING ($\beta = 1 - \alpha$) FOR TA* ($w = 8$), MEDIANS OVER THE TEST SCENARIOS, EVALUATED WITH THE DEFAULT COST MAP.

α	slope$_{95}$ [°]	rough.$_{95}$ [mm]	climb [m]	length [m]
0.00	14.2	18.4	3.52	88.9
0.25	9.2	19.5	3.27	90.0
0.50	8.8	20.6	3.25	92.3
0.75	8.5	21.5	3.24	93.7
1.00	8.3	22.6	3.15	94.4

G. Roll-aware extension

Table VI evaluates the roll angle along every path. At $\phi_{\max} = 20°$, A*, Theta* and A*-HC exceed the limit in 71%, 71% and 67% of the scenarios: the hard slope constraint alone does not prevent crossing side slopes. The isotropic TA*-S avoids almost all of this exposure because it avoids slopes altogether (3% of paths, 0.02 m on average), and TA*-S-R removes the rest at negligible cost: the median paired differences to TA*-S are -0.10° in maximum roll, +0.24 m in length and +0.09 kJ in energy, and the energy change relative to A* remains -6.6%. The roll term matters when the lateral limit is tighter than the slope limit, e.g., on slippery ground or with a high payload: at $\phi_{\max} = 10°$ the isotropic planner exceeds the limit in 56% of the scenarios, whereas TA*-S-R stays within it (the evaluation finds at most 0.10 m above it, maximum 10.6°, an artefact of resampling diagonal steps) and correctly reports infeasibility on 1 map. At $\phi_{\max} = 25° = \theta_{\max}$ the slope constraint already bounds roll, and no planner with the hard constraint exceeds it.

H. Computation time

Terrain awareness has a price in search effort: with $w > 0$ the Euclidean heuristic underestimates the true cost more, so TA* expands more nodes than A*. In our single-threaded Python implementation the median time on 100 m maps (200×200 cells) is 0.088 s for TA* and 0.116 s for TA*-S, against 0.021 s for A*; on 200 m maps it is 0.377 s and 0.477 s; the maximum over all scenarios is 0.80 s. All planners

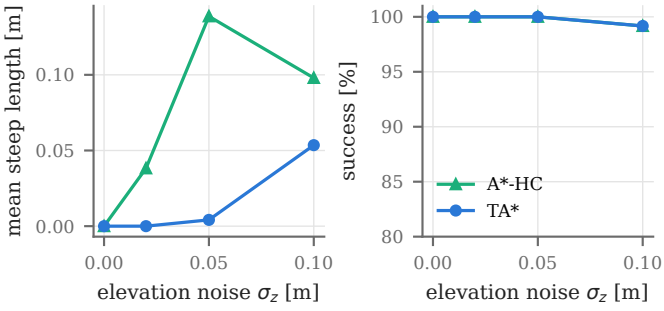


Fig. 3. Planning on a noisy elevation map, evaluation on the ground truth. Left: mean distance driven on slopes $> \theta_{\max}$. Right: success rate.

TABLE VI
ROLL EXPOSURE ON THE TEST SCENARIOS. EXPOSED: SHARE OF PATHS WITH ROLL ABOVE THE ROLL LIMIT $\phi_{\max}$ ANYWHERE; LEN.: MEAN DISTANCE ABOVE THE LIMIT; MAX: MAXIMUM ROLL; MAX, ΔL, ΔE (RELATIVE TO A*): MEDIANS; ALL AT $\phi_{\max} = 20°$ UNLESS STATED. TA*-S-R USES THE ROLL LIMIT IT IS EVALUATED WITH.

	Exposed [%] at $\phi_{\max}$							
Planner	10°	15°	20°	25°	len. [m]	max [°]	ΔL [%]	ΔE [%]
A*	97	92	71	48	5.88	24.7	0.0	0.0
Theta*	94	88	71	52	5.69	25.3	-5.7	-4.6
A*-HC	98	92	67	0	3.17	21.5	0.1	-1.4
TA*-S	56	17	3	0	0.02	10.4	3.0	-7.3
TA*-S-R	2	0	0	0	0.00	9.6	3.5	-6.6

scale similarly with map size; a compiled implementation or incremental replanning [6] would cut these times.

VI. Discussion and Limitations

Synthetic terrain. The terrains are procedural. They are designed to cover qualitatively different situations (unavoidable relief, a barrier with passes, local depressions, rough patches) and are scaled by slope, but they are not measured terrain. The pipeline accepts any elevation map; applying it to maps built with RTAB-Map [10] on our tracked UGV is the next step.

Attitude model. The roll-aware extension uses the terrain gradient under a point robot and an assumed roll limit; it ignores the vehicle's footprint, turning on slopes and dynamic effects. A heading state lattice and a footprint-based attitude estimate would refine it.

Energy and dynamics. The energy estimate ignores speed, slip, turning losses and regeneration, and no path is executed by a controller. The reported energy changes should therefore be read as relative indications, not predictions.

Implementation. Times are from a pure-Python implementation and are meant for comparison between planners, not as absolute real-time claims.

VII. Conclusion

A traversability cost normalised by the robot's own limits, embedded in A* with an admissible heuristic and followed by cost-aware shortcutting, removes all excursions onto slopes beyond the robot's limit that shortest-path planners make in most of our scenarios, lowers the mean traversability cost by a median of -39.6% and reduces climb and estimated energy, at a median distance overhead of a few percent. The terrain weight sets a clear distance–cost–energy trade-off that should be chosen per mission. We release the generator, planners, raw results and the analysis script that produces every number in this paper.

Code and Data Availability

All code, raw results and the analysis script are available at https://github.com/tako-1012/thesis_Bunker (directory `conference/`).

References

[1] P. Papadakis, "Terrain traversability analysis methods for unmanned ground vehicles: A survey," *Engineering Applications of Artificial Intelligence*, vol. 26, no. 4, pp. 1373–1385, 2013.
[2] D. C. Guastella and G. Muscato, "Learning-based methods of perception and navigation for ground vehicles in unstructured environments: A review," *Sensors*, vol. 21, no. 1, p. 73, 2021.
[3] S. B. Goldberg, M. W. Maimone, and L. Matthies, "Stereo vision and rover navigation software for planetary exploration," in *Proc. IEEE Aerospace Conference*, 2002.
[4] D. Ferguson and A. Stentz, "Using interpolation to improve path planning: The Field D* algorithm," *Journal of Field Robotics*, vol. 23, no. 2, pp. 79–101, 2006.
[5] P. E. Hart, N. J. Nilsson, and B. Raphael, "A formal basis for the heuristic determination of minimum cost paths," *IEEE Transactions on Systems Science and Cybernetics*, vol. 4, no. 2, pp. 100–107, 1968.
[6] S. Koenig and M. Likhachev, "D* Lite," in *Proc. AAAI Conference on Artificial Intelligence*, 2002, pp. 476–483.
[7] A. Nash, K. Daniel, S. Koenig, and A. Felner, "Theta*: Any-angle path planning on grids," in *Proc. AAAI Conference on Artificial Intelligence*, 2007, pp. 1177–1183.
[8] A. Nash, S. Koenig, and C. Tovey, "Lazy theta*: Any-angle path planning and path length analysis in 3D," in *Proc. AAAI Conference on Artificial Intelligence*, 2010.
[9] P. Fankhauser, M. Bloesch, and M. Hutter, "Probabilistic terrain mapping for mobile robots with uncertain localization," *IEEE Robotics and Automation Letters*, vol. 3, no. 4, pp. 3019–3026, 2018.
[10] M. Labbé and F. Michaud, "RTAB-Map as an open-source lidar and visual simultaneous localization and mapping library for large-scale and long-term online operation," *Journal of Field Robotics*, vol. 36, no. 2, pp. 416–446, 2019.
[11] M. Wermelinger, P. Fankhauser, R. Diethelm, P. Krüsi, R. Siegwart, and M. Hutter, "Navigation planning for legged robots in challenging terrain," in *Proc. IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2016.
[12] P. Krüsi, P. Furgale, M. Bosse, and R. Siegwart, "Driving on point clouds: Motion planning, trajectory optimization, and terrain assessment in generic nonplanar environments," *Journal of Field Robotics*, vol. 34, no. 5, pp. 940–984, 2017.
[13] N. Ganganath, C.-T. Cheng, and C. K. Tse, "A constraint-aware heuristic path planner for finding energy-efficient paths on uneven terrains," *IEEE Transactions on Industrial Informatics*, vol. 11, no. 3, pp. 601–611, 2015.
[14] F. Wilcoxon, "Individual comparisons by ranking methods," *Biometrics Bulletin*, vol. 1, no. 6, pp. 80–83, 1945.
[15] D. S. Kerby, "The simple difference formula: An approach to teaching nonparametric correlation," *Comprehensive Psychology*, vol. 3, 2014.
[16] S. Holm, "A simple sequentially rejective multiple test procedure," *Scandinavian Journal of Statistics*, vol. 6, no. 2, pp. 65–70, 1979.